

6.3 Classwork: Linear and Quadratic Functions

1. A straight line L passes through the points $A(-2, 5)$ and $B(4, -1)$.
 - (a) Find the gradient of L . [2]
 - (b) Find the equation of L in the form $y = mx + c$. [2]
 - (c) Find the x -intercept of L . [2]
 - (d) A second line L_2 is parallel to L and passes through the point $(0, 7)$. Find the equation of L_2 . [2]

2. Let $f(x) = x^2 - 4x - 12$.

- (a) Factorise $f(x)$. [2]
- (b) Hence find the x -intercepts of the graph of f . [1]
- (c) Find the coordinates of the vertex of the parabola. [3]
- (d) Write down the y -intercept. [1]
- (e) Write down the range of f . [1]

3. The line L_1 has equation $2x + y - 8 = 0$.
 - (a) Write the equation of L_1 in the form $y = mx + c$. [2]
 - (b) Write down the gradient and the y -intercept of L_1 . [1]
 - (c) The line L_2 is perpendicular to L_1 and passes through the point $(2, -1)$. Find the equation of L_2 in the form $ax + by + d = 0$, where $a, b, d \in \mathbb{Z}$. [3]
 - (d) Find the coordinates of the point of intersection of L_1 and L_2 . [3]

4. Let $g(x) = 2x^2 - 8x + 6$.

- Write $g(x)$ in the form $a(x - h)^2 + k$. [3]
- Write down the coordinates of the vertex of the parabola. [1]
- Find the x -intercepts of the graph of g . [2]
- The graph of g is reflected in the x -axis. Write the equation of the new graph in the form $y = ax^2 + bx + c$. [2]